

Arunabha Majumder

Mechanical Design Engineer | 3D CAD | Prototyping | Electromechanical Systems

Aalborg, Denmark | +45 71 62 24 00 | somrkmv1997@gmail.com |
linkedin.com/in/arunabha-majumder-681264107

PROFESSIONAL SUMMARY

Mechanical Design Engineer with a B.Tech in Mechanical Engineering, an M.Tech in Mechatronics and about three years of hands-on R&D design experience at Aalborg University. Designs, documents, prototypes and tests electromechanical products in 3D CAD (SolidWorks, Fusion 360), using FEA to iterate on stiffness, strength and weight before fabrication. Builds prototypes hands-on from machined and additively manufactured parts, integrates motors, sensors and DAQ, and analyses test data to solve design issues. Certified SolidWorks Associate (CSWA), fluent in English (C1) and experienced in R&D project teams in Denmark and India.

CORE SKILLS

- **3D CAD & Simulation:** SolidWorks (CSWA certified), Fusion 360, 3D modelling, engineering drawings, Finite Element Analysis (FEA), MATLAB/Simulink
- **Mechanical Design:** Mechanical Design, R&D Product Development, Design for Assembly (DFA), System Cost and Part-Count Reduction, Machined Parts, Additive Manufacturing / 3D Printing
- **Prototyping & Testing:** Specifying and building prototypes hands-on, Assembly, Custom Test Rigs, Experimental Validation, Root-Cause Analysis of Test Results
- **Electromechanical Systems:** Motor, Encoder, Force-Sensor and DAQ Integration, Embedded Control, Pneumatic Systems, Actuator Design
- **Communication:** English (C1, fluent written and spoken), technical writing (peer-reviewed publications), R&D project teamwork

PROFESSIONAL EXPERIENCE

Research Engineer – Mechanical Design & Actuator Development (PhD Project: VIEXO) | 2023 – Present

Aalborg University, Aalborg, Denmark

- Designed, documented and built a wearable electromechanical device from concept to tested hardware: SolidWorks 3D models and drawings, FEA-driven design iterations, machined and 3D-printed parts, hands-on assembly and custom automated test rigs.
- Integrated motor, encoder, force sensors and DAQ with embedded control into the functional prototype.
- Cut energy consumption by more than 25% by designing a hybrid actuation architecture that combines a parallel spring with a 6 Nm motor to offload gravity torque; awarded Best Paper, IFToMM ISRM 2026.
- Reduced system cost, mass and integration complexity of a 5-bar parallel robot by removing the external force/torque sensor, using a sensorless force-estimation method derived from the actuator torque–deflection characteristic.
- Validated analytical stiffness models against physical tests to below 10% RMS error, de-risking the mechanism before fabrication.
- Planned and ran a 12-participant validation study across 3 assistance modes and 2 load conditions, including instrumentation (EMG, motion capture), ethics protocol and data analysis; results support two journal submissions.

M.Tech Researcher – Actuator Systems (Master's Thesis) | 2020 – 2022

CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

- Fabricated pneumatic artificial muscle actuators in-house and integrated valves, pressure sensors and DAQ to characterise load–displacement behaviour and derive an empirical actuator model.
- Reduced dynamic position-tracking error to 0.3–0.78% with a neural-network-based gain-scheduled PID controller for a non-linear pneumatic actuator, outperforming the classical PID baseline under varying axial load.

Mechanical Engineering Capstone – Upper-Limb Rehabilitation Exoskeleton | 2018 – 2019

Siddaganga Institute of Technology, Tumakuru, India

- Led CAD design, fabrication and user testing of an upper-limb rehabilitation exoskeleton; won the Best Major Project Award and an INR 100,000 TATA Technologies Innovation Grant.

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present

Aalborg University, Department of Materials and Production, Aalborg, Denmark

M.Tech, Mechatronics | 2020 – 2022

AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

B.Tech, Mechanical Engineering | 2015 – 2019

Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

ENGINEERING PROJECTS

- **Go-Kart (2017):** Led end-to-end design and fabrication of a custom Go-Kart built from the ground up.
- **Efficycle (2018):** Worked in a multidisciplinary team to engineer a human-electric hybrid vehicle for the Efficycle competition.

CERTIFICATIONS & AWARDS

- Certified SolidWorks Associate (CSWA) – Mechanical Design
- Best Research Paper Award, 9th IFToMM International Symposium on Robotics and Mechatronics (2026)
- Best Major Project Award, Department of Mechanical Engineering (2019)
- TATA Technologies Innovation Award – INR 100,000 Grant (2019)

PUBLICATIONS

- Majumder, A., Wagner, J.W., Zhu, Y., Oliveira, A.S., and Bai, S. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." *Robotics and Mechatronics: ISRM 2026, MMS*, vol. 213, Springer. (Best Research Paper Award)
- Majumder, A., Sarkar, D., Chakraborty, S., Singh, A., Roy, S.S., and Arora, A. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." *IEEE CONECCT*, 2022.

TECHNICAL TOOLS

- **CAD & Manufacturing:** SolidWorks, Fusion 360, FEA, Bambu Lab and Ultimaker Cura (3D printing)
- **Software & Programming:** MATLAB/Simulink, Python, LabVIEW (basic), Arduino, LaTeX
- **Instrumentation & Testing:** Qualisys Motion Capture, Forsentec Force/Torque Sensors, Pneumatic Control, Custom Rig Fabrication

LANGUAGES

English (C1, fluent) | Bengali (Native) | Hindi (Conversational)

REFERENCES

Available upon request.